

UEI408: Analog Devices & Circuits				
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		3	1	2
Cr 4.5				
Course Objective: To enhance comprehension capabilities of students through understanding of analog electronic devices, BJT, FET and working of power supplies, amplifiers, oscillators and wave shaping circuits.				
Syllabus				
Bipolar Junction Transistors (BJT): CE configuration as two port network: -pi (II) Common-emitter Transistor Model, Hybrid-II conductances, The Hybrid-II Capacitances, The CE short-circuit current gain, Current gain with resistive load, Single-stage CE transistor amplifier response, The gain-bandwidth product, Emitter follower at high frequencies.				
Transistors Amplifier: Multistage Amplifier: Classification of amplifiers, Analysis of transistor amplifier using h – parameter, Frequency response of single stage RC coupled CE amplifier, Effect of an emitter Bypass capacitor on low-frequency response, Feedback Amplifiers: Classification of amplifiers, Positive and negative feedback, Effect of feedback on gain, input and output impedances. h _{ie} parameters, h _{ie} parameter equivalent circuit. The Hybrid				
Field-Effect Transistors (FET): Structure and working of JFET and MOSFET, output and transfer characteristics, FET as voltage variable resistor and MOSFET as a switch. Biasing the FET, The FET small signal model, the low-frequency common-source and common-drain amplifiers.				
Oscillator Circuits: Condition for sustained oscillation, R-C phase shift, Wien Bridge, Hartley, Colpitts and Crystal Oscillators, Frequency stability, Multivibrators, High pass and low pass filters using R-C Circuits and R-L, R-L-C Circuits, Attenuators, Clamping Circuit theorem, Schmitt Trigger, Comparator.				
Signal Conditioning: Operational Amplifiers: application in instrumentation, Charge amplifier, Carrier amplifier, Introduction to active filters, Classification, Butterworth, Chebyshev, Couir filters, First order, Second order and higher order filters, Voltage to frequency and frequency to voltage converters.				
Laboratory Work: Transistor circuits using BJT and FET, oscillators, filters.				
Course Learning Objectives (CLO) The students will be able to: 1. Differentiate between different of diodes based on their working principle. 2. Elucidate the working principle of BJT and FET				

3. Analyse transistor amplifier using h-model and analyse the effect of feedback on amplifiers. 4. Design the oscillator circuit. 5. Elucidate the concept of active filters.
Text Books 1. Electronic Devices and Circuit Theory, Boylestad R. L., Pearson Education (2007) 9 th ed. 2. Integrated Electronics, Millman, J. and Halkias, C.C., Tata McGraw Hill (2006).
Reference Books 1. Neamen, Donald A., Electronic Circuit Analysis and Design, McGraw Hill (2006) 3 rd ed. 2. Sedra A. S. and Smith K. C., Microelectronic Circuits, Oxford University Press (2006) 5 th ed.

Evaluation Scheme	
Evaluation Elements	Weightage %
Mid Semester Test (MST)	25-30
End Semester Examination (ESE)	40-45
Sessional (may include Assignment, Sessional (Includes Regular Lab assessment) and Quizzes Project (Including report, presentation etc.)	30